

Targeted Deep Survival Contrasts: Valid Inference for Treatment-Specific Survival Benefit with Neural Networks

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Abstract

Neural survival models are increasingly asked to support *counterfactual* claims—how much a treatment would change survival in a population—rather than only prognostic risk scores. Answering such questions from observational data requires valid inference for treatment-specific survival contrasts under confounding and covariate-dependent censoring, targets for which standard deep survival estimators are biased and provide no honest uncertainty. We propose *Targeted Deep Survival Contrasts* (TDSC), which extends Targeted Deep Architectures (TDA)—targeted maximum likelihood estimation embedded in a network’s weight space—to the full vector of treatment-specific survival curves $\{S_1(t), S_0(t)\}$ over a time grid, and hence to the benefit curve $S_1(t) - S_0(t)$ and the restricted mean survival time (RMST) difference. A single universal targeting path, one ridge projection of the stacked efficient influence functions onto closed-form last-layer gradients per iteration, simultaneously solves the projected estimating equations for all coordinates; a one-step residual “top-up” converts the plug-in into a doubly robust estimator of the unrestricted target; and a multiplier bootstrap yields simultaneous confidence bands for the benefit curve. We prove joint asymptotic linearity, band validity, and double robustness of the top-up for a cross-fitted variant requiring no Donsker conditions. Across seven Monte Carlo banks with confounded treatment, sign-varying effect heterogeneity, and dependent censoring, the TDSC plug-in attains nominal pointwise and simultaneous coverage (94.8% and 95.0% in the main bank) with 35% lower MSE and narrower intervals than a per-timepoint one-step (AIPCW) built from the same nuisance fits, an advantage that persists across $n \in \{500, \dots, 2000\}$, a nonlinear design, and two orders of magnitude of ridge penalty. The two estimators divide the labor cleanly: under a badly wrong outcome model the plug-in faithfully tracks its *working* parameter and its intervals fail (44% coverage), while the top-up restores nominal inference for the unrestricted causal target (94–95%)—and in-sample diagnostics separate the two regimes replication by replication.

1 Introduction

Neural networks are now standard tools for individual-level survival prediction from rich clinical inputs, including digital pathology and other high-dimensional modalities. Clinical decisions, however, hinge on *contrasts*: how survival would differ under one treatment versus another. Estimating such contrasts from observational data raises two familiar obstacles—confounding and censoring that depends on treatment and covariates—and a less appreciated third: a network trained for predictive loss is a biased, inefficient estimator of any low-dimensional causal functional [3, 10], and its naive standard errors are invalid.

Semiparametric theory offers the remedy: debias a flexible initial fit by solving the efficient influence function (EIF) equation, via one-step/AIPCW corrections [3, 6] or targeted maximum

likelihood estimation (TMLE) [13]. For survival targets these tools are mature [8, 1, 9], but standard implementations correct *outside* the model: the one-step adds the empirical mean of the EIF to a plug-in, and classical TMLE fluctuates the output distribution after training. Both become awkward when the target is a whole *vector*—two survival curves on a fine grid—where per-coordinate corrections need not respect monotonicity and can push estimates off the model space. Targeted Deep Architectures (TDA) [7] takes a different route: it embeds the TMLE fluctuation *inside* the network. A small targeting subset of weights (here, the final linear layers of the outcome heads) is updated along the projection of the EIF onto the per-sample gradients of the network’s own training loss, so the fitted network itself becomes the debiased estimator—its outputs remain genuine survival curves, monotone by construction, and no post-hoc correction sits outside the model. TDA was demonstrated for the average treatment effect and for the marginal survival curve without treatment; the clinically decisive object—the causal contrast of treatment-specific survival curves under confounding and dependent censoring—was not addressed.

This paper closes that gap. TDA already supplies the machinery for many targets at once—its universal path merges per-target directions obtained by projecting stacked EIFs onto the working score span—and we take that machinery as given. What the survival contrast adds is everything around it: a $2K$ -dimensional target whose EIFs combine treatment and censoring weights under confounding and dependent censoring; closed-form last-layer gradients that make each merged update a single ridge regression, with no per-sample automatic differentiation; and, running alongside, a deliberately two-part inferential design. The targeted *plug-in* solves the estimating equations that the network’s working submodel can actually solve—the projected ones—and earns adaptive-efficiency gains from that restriction; a one-step residual *top-up* then adds back whatever full-EIF bias the restriction leaves behind, recovering a regular, doubly robust estimator of the unrestricted causal target. The two estimators answer different questions, their variances are priced by different influence functions, and much of the paper is devoted to keeping that distinction honest—theoretically, and in simulations designed to make each estimator fail where it should.

Related work. Neural estimators of treatment-specific survival curves exist—notably SurvITE [5], which targets the conditional counterpart of our estimand with balanced discrete-time hazard networks, and counterfactual time-to-event models [2]—but they provide point estimates without valid confidence statements; that is the gap we fill. Conversely, survival TMLEs with simultaneous inference exist [12, 1, 9], including universal one-dimensional least-favorable submodels that target an entire survival curve at once; those paths are constructed *analytically*, per estimand and likelihood, whereas TDA’s is obtained generically from the architecture’s score gradients. TDSC’s contribution on this axis is the contrast-specific instantiation of that generic path—the confounded, censored two-arm stack with closed-form last-layer gradients—together with the inference package built on top of it: the working-model versus full-EIF variance split, the doubly robust residual top-up, simultaneous bands, and cross-fitted theory. Causal survival forests [4] give honest inference for conditional effects at fixed horizons rather than the full population benefit curve.

Contributions. We propose *Targeted Deep Survival Contrasts* (TDSC), with four contributions:

1. the target vector $\Psi = (S_1(t_1), \dots, S_1(t_K), S_0(t_1), \dots, S_0(t_K))$ with its stacked EIFs combining treatment and censoring weights (Section 2);
2. closed-form per-sample last-layer gradients that make the universal targeting update for all $2K$ coordinates a single ridge regression per iteration (Algorithm 1), with multiplier-bootstrap sup- t bands for the benefit curve (Section 3);

3. formal guarantees—joint asymptotic linearity, band validity, and a residual top-up that provably restores point-estimate double robustness—for a cross-fitted variant in which sample splitting replaces the Donsker-class conditions of the underlying TDA/ADML theory (resolving, for the top-up, the cross-fitted extension left open in 7), together with a design rule (and a documented failure mode) for *where* the high-dimensional targeting step may be solved (Sections 3–4);
4. a simulation program—a unified main bank, nuisance-misspecification and sample-size grids, a cross-fitting comparison, and head-to-head external comparators (output-space survival TMLE, causal survival forests)—that decomposes the mechanism behind TDSC’s gains and maps its operating envelope (Section 5).

2 Setup, estimands, and efficient influence functions

Data. We observe n i.i.d. copies of $O = (X, A, \tilde{T}, \Delta)$: covariates $X \in \mathbb{R}^d$, binary treatment $A \in \{0, 1\}$, discrete follow-up time $\tilde{T} \in \{1, \dots, K\}$, event indicator Δ . Time is coarsened into K bins; within a bin, events are recorded before censoring, and subjects surviving bin K are administratively censored. Let $Y(k) = \mathbf{1}\{\tilde{T} \geq k\}$ (at risk) and $dN(k) = \mathbf{1}\{\tilde{T} = k, \Delta = 1\}$ (event), and write $P_n f = n^{-1} \sum_i f(O_i)$. The likelihood factorizes through the propensity $g(X) = P(A = 1 \mid X)$, the event hazard $h(k \mid a, x)$, and the censoring hazard $h_c(k \mid a, x)$, with conditional survival $S(t \mid a, x) = \prod_{k \leq t} \{1 - h(k \mid a, x)\}$ and censoring survival S_c defined analogously.

Estimands. Under consistency, no unmeasured confounding, coarsening-at-random censoring (censoring may depend on A and X but not further on the event time), and positivity for treatment and censoring, the treatment-specific curves $S_a(t) = \mathbb{E}_X[S(t \mid a, X)]$, $a \in \{0, 1\}$, $t \in \{1, \dots, K\}$, are identified. Our target is the $2K$ -vector $\Psi = (S_1(\cdot), S_0(\cdot))$; the *benefit curve* $\beta(t) = S_1(t) - S_0(t)$ and the RMST difference $\Delta_{\text{RMST}} = \sum_{t=1}^K \beta(t)$ (bin width one; the contrast in expected bins survived through horizon K , i.e. in $\mathbb{E}[\min(T - 1, K)]$ —the conventional $\mathbb{E}[\min(T, K)]$ contrast would be $\sum_{t=1}^{K-1} \beta(t)$) are linear in Ψ , so their influence functions follow by linearity. These are the quantities a treatment decision actually turns on: $\beta(t)$ is the absolute survival gain the population would realize at horizon t if everyone were treated versus no one, and Δ_{RMST} summarizes it as extra expected survival time. Targeting the whole vector at once, rather than one horizon at a time, is what makes simultaneous inference over the curve possible.

Efficient influence functions. For each coordinate $S_a(t)$ the EIF at P is the one underlying the randomized-trial TMLE of Moore and van der Laan [8], extended to the observational setting (estimated propensity in place of randomization; censoring dependent on treatment and covariates):

$$D_{a,t}(O) = - \sum_{k=1}^t \frac{\mathbf{1}\{A = a\}}{g_a(X) S_c(k-1 \mid a, X)} \frac{S(t \mid a, X)}{S(k \mid a, X)} (dN(k) - Y(k) h(k \mid a, X)) + S(t \mid a, X) - S_a(t), \quad (1)$$

with $g_1 = g$, $g_0 = 1 - g$. Each piece has a role worth naming. The summand is a martingale residual—the within-bin “surprise” of an event relative to the fitted hazard—reweighted by the inverse probability of receiving arm a and remaining uncensored through bin $k - 1$, so that the observed arm- a , still-at-risk subjects stand in for the full population; the factor $S(t \mid a, X)/S(k \mid a, X)$ propagates a hazard error at bin k into its effect on survival at t ;¹ and the final terms center the

¹Ratios $S(t)/S(k)$ denote the tail product $\prod_{j=k+1}^t \{1 - h(j \mid a, x)\}$, well defined even where $S(k) = 0$; the implementation additionally floors conditional survival (Appendix A).

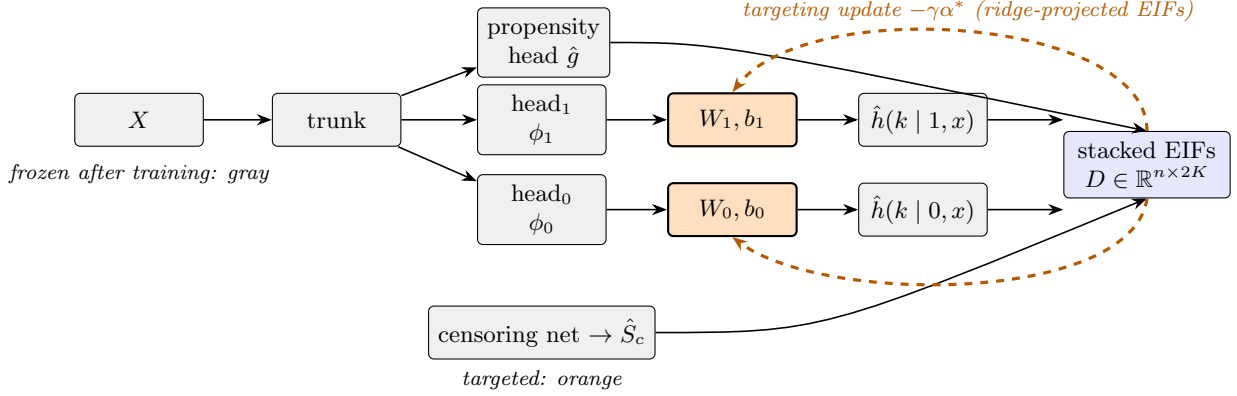


Figure 1: TDSC pipeline. A DragonNet-style survival network produces per-arm hazards; the propensity head and a separate censoring network supply the weights inside the stacked EIFs. Only the final linear layers (W_a, b_a) are updated, along the universal direction obtained by ridge-projecting all $2K$ EIFs onto their closed-form per-sample gradients; everything else stays frozen.

plug-in. Stacking (1) over a and t gives the $n \times 2K$ influence matrix D (throughout, D denotes the influence matrix evaluated at the current fitted nuisances; hats distinguish fitted from population objects only where both appear) whose column means $P_n D_{a,t}$ are the first-order bias terms of the plug-in. The targeting step of Section 3 drives the *projections* of these means onto the working score span to zero, and the residual top-up removes whatever the projection leaves behind.

3 Targeted Deep Survival Contrasts

Architecture and initial fit. We use a DragonNet-style network [10] adapted to discrete-time survival: a shared trunk feeding a propensity head for g and two outcome heads producing hazard logits $\{h(k | a, x)\}_{k=1}^K$ for $a \in \{0, 1\}$, trained on the observed-arm discrete-time negative log-likelihood (masked per-bin binary cross-entropy) plus a propensity term, with early stopping. A separate network fits the censoring hazard. Any backbone—including a frozen foundation-model encoder with a trained head—can replace the trunk. Full architecture and hyperparameters are in Appendix A.

Targeting submodel and closed-form gradients. Following TDA, the targeting parameters ϑ are the final linear layers of the two outcome heads, $p = 2K(F+1)$ parameters for head width F . Because those layers are linear in the head features $\phi_a(X)$ and the loss is a per-bin Bernoulli likelihood, the per-sample gradient is available in closed form:

$$\frac{\partial \ell_i}{\partial W_a[k, \cdot]} = \mathbf{1}\{A_i = a\} Y_i(k) (h(k | a, X_i) - dN_i(k)) \phi_a(X_i), \quad (2)$$

and analogously for biases, giving the $n \times p$ score matrix G in one vectorized pass with no per-sample automatic differentiation.

Universal targeting update. Each iteration solves one ridge system for all $2K$ targets at once,

$$\hat{\alpha}_\lambda = (G^\top G + \lambda I)^{-1} G^\top D,$$

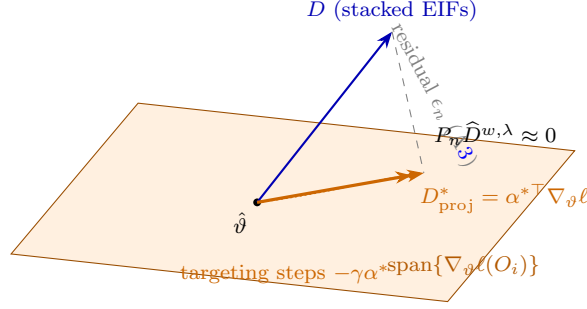


Figure 2: Geometry of the targeting update. The stacked nonparametric EIFs D are projected onto the span of the per-sample loss gradients of the targeting parameters; TDSC steps ϑ along the projected direction until the *projected* estimating equations hold. The observable residual ϵ_n (Assumption 3) measures how much of the nonparametric EIF lies outside the working score span—the gap between the two influence-function geometries and the potential efficiency gain from the restriction, not the parameter-value bias, which the oracle-bias conditions govern separately—and the full-EIF means it leaves behind are exactly what the one-step residual top-up adds back.

giving a direction $\hat{\alpha}_{\lambda,j}$ in weight space for each coordinate j and the projected bias terms $d_j = P_n[G\hat{\alpha}_{\lambda,j}]$. The ridge is not decoration: the targeting layers have $p \approx 4,000$ parameters against $n = 1,000$ observations, so the unpenalized projection is ill-posed, and λ (scaled to the mean diagonal of $G^\top G$) is statistically active. One deviation from TDA as published: its projection fits an unpenalized intercept per coordinate (equivalently, centers scores and EIFs before the solve); we solve the intercept-free system above. Away from a full-sample optimum the score columns need not be empirically centered, so the two projections can differ by rank-one mean terms; the EIF columns are centered by construction, and near the trained optimum the score means are small, so the discrepancy was negligible here. Writing $\hat{D}^{w,\lambda} = G\hat{\alpha}_\lambda$ for the resulting ridge estimate of the working-submodel influence functions (Section 4 separates it from its population counterpart D_n^w), the $2K$ per-target directions are merged into one universal direction, $\alpha^* = \sum_j (d_j/\|d\|_2) \hat{\alpha}_{\lambda,j}$, following the multi-target merge rule of Li et al. [7]: each coordinate pulls the weights along its own direction, in proportion to how biased it currently is. The update $\vartheta \leftarrow \vartheta - \gamma \alpha^*$ uses a backtracking line search on $\sum_j (P_n \hat{D}_j^{w,\lambda})^2$ ($\hat{\alpha}_\lambda$ held fixed within the search); after each accepted step the hazards, influence matrix, and projection are recomputed, while g and S_c stay fixed.

The stopping criterion is the projected one: iteration ends when every coordinate satisfies

$$|P_n \hat{D}_j^{w,\lambda}| \leq \widehat{\text{sd}}(\hat{D}_j^{w,\lambda})/(\sqrt{n} \log n).$$

The $\text{sd}/(\sqrt{n} \log n)$ tolerance follows Rytgaard and van der Laan [9] via TDA; relative to TDA’s aggregate criterion—one norm across all targets—we require every coordinate to clear its own tolerance, which is the form coordinatewise inference needs. These are the estimating equations of the *working submodel*—the equations a restricted parameterization can actually solve. The *full*-EIF means $P_n D_j$ need not vanish inside a restricted submodel, and demanding that they do would ask the final layers for something they cannot deliver. Instead the full-EIF means are tracked, and adding them back as a one-step residual top-up, $\hat{\Psi} + P_n D$, converts the plug-in into an estimator of the unrestricted target (Proposition 3). This division—solve what the submodel can solve, correct linearly for the rest—is the organizing idea of the method.

Two diagnostics come for free. The first is the top-up magnitude itself, $\max_j |P_n D_j|$: how much full-EIF bias the restricted step left behind. The second is the relative projection residual $\|D - \hat{D}^{w,\lambda}\|/\|D\|$, whose interpretation Section 4 makes precise: it measures how much of the

Algorithm 1 TDSC universal targeting (no-split form)

Require: trained hazard/propensity network and censoring network; targeting parameters ϑ (final layers of the outcome heads); ridge scale λ

- 1: fix weights $\hat{g}, \hat{S}_c$; compute hazards $\hat{h}_a$, survival curves, and stacked full EIFs $D \in \mathbb{R}^{n \times 2K}$ via (1)
- 2: **loop**
- 3: $G \leftarrow$ per-sample loss gradients w.r.t. ϑ ▷ closed form (2); no autograd
- 4: $\hat{\alpha}_\lambda \leftarrow (G^\top G + \lambda I)^{-1} G^\top D$; $\hat{D}^{w,\lambda} \leftarrow G \hat{\alpha}_\lambda$ ▷ ridge estimate of the working EIFs
- 5: **if** $|P_n \hat{D}_j^{w,\lambda}| \leq \widehat{\text{sd}}(\hat{D}_j^{w,\lambda})/(\sqrt{n} \log n)$ for all j **then break** ▷ projected criterion
- 6: $d_j \leftarrow P_n \hat{D}_j^{w,\lambda}$; $\alpha^* \leftarrow \sum_j (d_j / \|d\|_2) \hat{\alpha}_{\lambda,j}$ ▷ universal direction
- 7: backtracking line search: $\vartheta \leftarrow \vartheta - \gamma \alpha^*$, accepting only if $\sum_j (P_n \hat{D}_j^{w,\lambda})^2$ falls ($\hat{\alpha}_\lambda$ fixed within the search); **break** if no γ improves
- 8: recompute $\hat{h}_a$, curves, and D at the updated ϑ
- 9: **end loop**
- 10: **return** plug-ins $\hat{\Psi}_j$; full-residual top-up $\hat{\Psi}_j + P_n D_j$; plug-in SEs $\widehat{\text{sd}}(\hat{D}_j^{w,\lambda})/\sqrt{n}$, top-up SEs $\widehat{\text{sd}}(D_j)/\sqrt{n}$; multiplier-bootstrap band; diagnostics ($\|D - \hat{D}^{w,\lambda}\|/\|D\|$, $\max_j |P_n D_j|$)

nonparametric EIF lies outside the working score span—the gap between the two influence-function *geometries*, hence the potential efficiency gain from the restriction—not the parameter-value bias $\Psi_n(P_0) - \Psi(P_0)$, which is governed separately by the oracle-bias conditions. Section 5 shows both diagnostics jumping sharply exactly when the working submodel is badly wrong. Finally, because the update moves weights rather than outputs, the plug-in estimator is by construction a pair of monotone survival curves.

Cross-fitted variant. To replace the Donsker-type conditions of the theory below with sample splitting, TDSC admits a V -fold cross-fitted form: for each fold, the nuisance networks are trained *and targeted* on the complement; the held-out fold then supplies evaluation—plug-in curves, influence functions at the targeted fit, and an exact residual correction $P_{n,v}[D]$ so the EIF equations hold exactly on held-out data (by Proposition 3, a cross-fitted one-step at the targeted nuisances). Fold contributions are averaged; the pooled per-observation influence matrix supplies variance and bands.

Why not target fold-wise on the held-out folds? A tempting alternative solves the EIF equations with the full targeting submodel *separately on each held-out fold*. This fold-wise high-dimensional targeting fails badly here, and the mechanism is instructive. It is *not* what classical CV-TMLE [11] does: there, a single *low-dimensional* fluctuation is fit by pooling the cross-validated loss over all validation folds— $O(1)$ parameters against n observations—which is exactly why it is safe (the faithful CV-TMLE analogue here would pool the $2K$ one-dimensional fluctuations of Section 5’s output-space TMLE across folds, which we have not run). TDA’s submodel, by contrast, is the network’s final layers ($p \approx 4,000$ here), so on a fold of $n/V \approx 200$ observations the gradient matrix has rank at most 200 and the fold’s empirical EIF equations can be solved nearly exactly—by fitting the fold’s *sampling noise*, whose order $\text{sd}(D)/\sqrt{n/V}$ rivals the genuine first-order bias the step is meant to remove. The resulting hazard perturbations pass through the nonlinear map $h \mapsto \prod_k (1 - h(k))$, rectifying symmetric fold-level fluctuation into systematic bias, and the fold-fitted targeting layers reintroduce exactly the empirical-process term $(P_n - P_0)(\hat{D} - D_0)$ that cross-fitting is meant to eliminate—now concentrated on the smallest split. Empirically this variant gave 14-fold larger benefit-curve bias than the no-split estimator, 70.5% pointwise and 42% simultaneous coverage (100 replications; store preserved in the repository). The design rule:

the dimension of the data-adaptive targeting step must be small relative to the split on which its equations are solved, so targeting belongs with training on the large split, and the held-out fold should receive only evaluation and the linear residual correction. Section 5 compares the no-split and cross-fitted forms empirically.

Inference and simultaneous bands. Two variance prescriptions correspond to the two inferential targets of Section 4: the *top-up* estimator (unrestricted target) uses the full-EIF standard errors $\widehat{\text{sd}}(D_j)/\sqrt{n}$, while the *plug-in* (working target) uses the working-submodel standard errors $\widehat{\text{sd}}(\widehat{D}_j^{w,\lambda})/\sqrt{n}$, both at the final fit; contrasts inherit influence functions by linearity. For the benefit curve we report a sup- t band over the prespecified grid: with i.i.d. standard normal multipliers ξ_i and influence values centered at their empirical mean $\bar{D}_{\beta,t} = P_n D_{\beta,t}$, the 95% quantile c^* of $\max_t |n^{-1} \sum_i \xi_i \{D_{\beta,t}(O_i) - \bar{D}_{\beta,t}\}|/\widehat{\text{se}}_t$ over bootstrap draws gives $\hat{\beta}(t) \pm c^* \widehat{\text{se}}_t$ (c^* floored at 1.96, a conservative choice); here $D_{\beta,t}$ denotes the influence values matching the chosen variance prescription—full-EIF for the top-up band, working-model for the plug-in band.

4 Theory

TDSC is TDA applied to a $2K$ -dimensional pathwise-differentiable target, and its guarantees specialize the working-parameter and original-parameter theorems of Li et al. [7] and adaptive debiased machine learning (ADML) [14]. We state the specialization for the *cross-fitted* implementation (Section 3; V -fold, nuisances trained and targeted out-of-fold, evaluation with exact residual correction on the held-out fold), in which sample splitting replaces those results’ Donsker-class conditions—fully so for the top-up estimator (Theorem 1, resolving the cross-fitted extension left open in TDA); the plug-in still imports a Donsker-type condition on the training folds, where its projected score equations are solved (Proposition 1).

Three influence-function objects must be kept distinct, because the paper’s two estimators are asymptotically linear with *different* ones:

- D_0^{np} , the $2K$ -vector of *nonparametric* EIFs (1) at P_0 : the influence function of the unrestricted target $\Psi(P_0)$, and the one the top-up estimator attains.
- D_n^w , the *working-model* EIF of the working parameter $\Psi_n(P_0) = \Psi(\Pi_n P_0)$ of the data-adaptive submodel $M_n \subseteq \mathcal{M}$ induced by partially unfreezing ϑ . Because M_n restricts only the hazards while the covariate marginal remains nonparametric, D_n^w retains the centered plug-in component $S_{\Pi_n P_0}(t | a, X) - \Psi_{n,j}(P_0)$ intact and projects only the *martingale* component of D_0^{np} onto the closure of the working score span; the hazard scores are conditionally centered given (A, X) , so that span is orthogonal to functions of X alone, and projecting the full EIF onto it would wrongly delete the covariate-marginal variance.² Under ADML, D_n^w converges to an *oracle*-submodel EIF D^O , and the plug-in is asymptotically linear with D^O .
- $\widehat{D}^{w,\lambda} = G\widehat{\alpha}_\lambda$, the ridge-regularized *empirical* counterpart the algorithm actually computes. The algorithm ridge-projects the full influence matrix rather than its martingale component alone; the centering component has empirical mean zero, so the stopping rule is unaffected, and with $p > n$ the near-interpolating fit retains most of that component in $\widehat{D}^{w,\lambda}$ (compare the working SE 0.032 to the full-EIF SE 0.034 in Table 1), at the price of making the empirical working SE’s target mildly ridge-dependent.

²Strictly, the EIF of Ψ_n in M_n is Hessian-based; the L^2 score-span projection replaces the Hessian by the score covariance. Because the loss is a per-bin Bernoulli likelihood, Bartlett’s identity makes the two coincide when M_n contains P_0 and, under Assumption 3, in the oracle limit.

Conflating the first two is what a careless reading invites: the entire variance story lives in the gap between D_0^{np} and D^O . For the third object, the assumption splits by use: for the plug-in standard errors, $\|\hat{D}^{w,\lambda} - D_n^w\|_{P_0} = o_p(1)$ suffices; for the stopping rule to certify the working-model estimating equations at the $o_p(n^{-1/2})$ level Proposition 1 requires, we additionally assume $P_n(\hat{D}^{w,\lambda} - D_n^w) = o_p(n^{-1/2})$ —the analogue of TDA’s approximation condition, whose empirical-process half cross-fitting controls and whose drift half is bounded by the product of the ridge-approximation and initial-estimator rates. A sufficient condition requires the ridge to vanish relative to the *nonzero spectrum* of the empirical score covariance—for example, under a uniform lower bound on its nonzero eigenvalues, $\lambda_n/n \rightarrow 0$ suffices; a condition on the mean diagonal alone does not control nearly unidentified score directions. Our simulations fix the relative ridge at 0.01 as a finite-sample stabilizer; this choice implements, but is not itself covered by, the asymptotic condition (Section 5 varies it over two orders of magnitude without material effect).

Assumption 1 (Identification and positivity). *Consistency, no unmeasured confounding, coarsening-at-random censoring; and $\delta \leq g_0(X) \leq 1 - \delta$, $S_{c,0}(K - 1 \mid a, X) \geq \delta$ a.s. for some $\delta > 0$.*

Assumption 2 (Nuisance rates). *For $f = f(k, a, x)$ write $\|f\|_{P_0}^2 = \sum_a \sum_{k=1}^K \mathbb{E}_{P_0} f(k, a, X)^2$. The cross-fitted weight estimators and the targeted per-fold hazard estimators are each consistent in $\|\cdot\|_{P_0}$, respect the bounds of Assumption 1, and satisfy the product rate $\|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ (a sufficient condition for the sharper per-(a, t) sums of per-bin products appearing in the remainder). Consistency and the fixed truncation levels of Appendix A are compatible only when the truncation sits inside the positivity bound ($\delta \geq 0.05$ here); the near-positivity design deliberately violates this, and Section 5 notes the induced bias.*

Assumption 3 (Gradient coverage—plug-in story only). *The projection residual $\epsilon_{n,j} = D_{0,j}^{\text{np}} - D_{n,j}^w$ stabilizes, in two senses: D_n^w converges in $L^2(P_0)$ to an oracle-model EIF D^O , and $(P_n - P_0)(D_n^w - D^O) = o_p(n^{-1/2})$ (under cross-fitting the second follows from the first by conditional Chebyshev, since M_n is selected independently of the evaluation fold). This assumption is not required for Theorem 1. It governs only the plug-in story (Proposition 1), and the size of $\|\epsilon_n\|$ is a dial, not a defect—but note carefully what it measures: the residual is a statement about influence-function geometry, not parameter values. Even when the oracle submodel contains P_0 (so $\Psi(\Pi_0 P_0) = \Psi(P_0)$ exactly and there is no approximation bias), D^O can remain a strict projection of D_0^{np} onto a smaller tangent space, with $\text{Var}(D^O) < \text{Var}(D_0^{\text{np}})$ —the same parameter with a lower local efficiency bound. That is the ADML superefficiency mechanism. Approximation bias $\Psi_n(P_0) - \Psi(P_0)$ is governed separately by TDA’s oracle-bias conditions. The empirical residual $\|D - \hat{D}^{w,\lambda}\|/\|D\|$ estimates the geometric gap.*

Theorem 1 (Joint asymptotic linearity and efficiency of the top-up estimator). *The full-residual top-up estimator is the cross-fitted AIPCW one-step evaluated at the targeted nuisances (Proposition 3); no separate “EIF equations solved” condition is needed. Under Assumptions 1 and 2 alone (Assumption 3 not required), $\hat{\Psi}^+$ satisfies jointly in $\mathbb{R}^{2K}$ $\hat{\Psi}^+ - \Psi(P_0) = P_n D_0^{\text{np}} + o_p(n^{-1/2})$, so $\sqrt{n}(\hat{\Psi}^+ - \Psi(P_0)) \rightsquigarrow N(0, \Sigma_0)$ with $\Sigma_0 = P_0 D_0^{\text{np}} (D_0^{\text{np}})^\top$: the estimator is regular and attains the nonparametric efficiency bound. Targeting enters only through Assumption 2—it must not degrade, and is designed to improve, the targeted hazards’ rates—so the theorem claims no asymptotic variance advantage for weight-space targeting; any benefit is finite-sample, through better nuisances.*

Proposition 1 (Cross-fitted plug-in: working parameter and superefficiency). *The cross-fitted TDSC plug-in (no top-up) is instead an ADML-type estimator of the working parameter $\Psi_n(P_0) = \Psi(\Pi_n P_0)$: under Assumption 3 and the conditions of the working-parameter and original-parameter theorems*

of Li et al. [7]—including their Donsker-type condition, imported here on the training folds, where the projected score equations are solved (cross-fitting removes that condition for the top-up but not for the plug-in), and their oracle-bias conditions under which $\Psi_n(P_0)$ may replace $\Psi(P_0)$ —it is asymptotically linear with the oracle EIF D^O . If in addition the oracle submodel contains P_0 ($\Pi_0 P_0 = P_0$), then $\text{Var}(D^O) \preceq \Sigma_0$ in the Loewner order, with improvement in direction v exactly equal to $\text{Var}(v^\top (D_0^{\text{np}} - D^O))$, strict wherever that limiting residual variance is nonzero (superefficiency, via the likelihood-based oracle-submodel extension of TDA and ADML—our loss is a per-bin Bernoulli likelihood—with the usual price: nonregularity for the unrestricted target under local alternatives outside the oracle model). Its inference must therefore use the working-model standard errors $\widehat{\text{sd}}(\widehat{D}_j^{w,\lambda})/\sqrt{n}$; pairing the plug-in with full-EIF standard errors is conservative. Both signatures—lower MSE than Theorem 1’s estimator, and over-coverage under full-EIF variance—appear in Section 5.

Corollary 1 (Contrasts). *The benefit curve $\hat{\beta}(\cdot)$ and $\widehat{\Delta}_{\text{RMST}}$ are asymptotically linear by linearity of both maps, with influence functions $D_{\beta,t}^{\text{np}} = D_{1,t}^{\text{np}} - D_{0,t}^{\text{np}}$ and $\sum_t D_{\beta,t}^{\text{np}}$ for the top-up estimator of Theorem 1, and $D_{\beta,t}^O = D_{1,t}^O - D_{0,t}^O$ and $\sum_t D_{\beta,t}^O$ for the plug-in of Proposition 1.*

Proposition 2 (Simultaneous band). *Under the conditions of Theorem 1 and the nondegeneracy $\min_t \text{Var}(D_{\beta,t}^{\text{np}}) > 0$ (immediate for a nondegenerate event hazard), the Gaussian-multiplier estimate of the 95% quantile of $\max_t |\mathbb{G}_t|/\sigma_t$, for $\mathbb{G}$ the limiting Gaussian of Corollary 1, is consistent, and the sup- t band covers the benefit curve over the prespecified grid with asymptotic probability at least 0.95 (the $c^* \geq 1.96$ floor can only enlarge it). Under Proposition 1 plus consistent estimation of $\text{Cov}(D^O)$ by the empirical working-model IFs (and the corresponding nondegeneracy), the same multiplier construction applies to the plug-in; in each case the multiplier statistic must be built from the same influence values as the chosen variance prescription.*

Proposition 3 (Top-up restores double robustness). *Define the top-up estimator with the full residual, $\widehat{\Psi}^+ = \widehat{\Psi} + P_n \widehat{D}$; it equals the AIPCW one-step evaluated at the targeted nuisances. Consequently—for the cross-fitted form, under Assumption 1 and convergence of the estimated nuisances to some limits respecting the bounds of Assumption 1 (enforced in practice by the truncation of Appendix A)— $\widehat{\Psi}^+$ is consistent for $\Psi(P_0)$ if either the hazard limit equals h_0 or the weight limits equal $(g_0, S_{c,0})$ jointly, regardless of whether targeting converged. The no-split form additionally requires a uniform law of large numbers for the estimated-EIF class, which cross-fitting renders unnecessary.*

Proof sketches are in Appendix B; they specialize the ADML arguments using the exact second-order remainder of the survival EIF (a telescoping identity), with cross-fitting controlling the empirical-process term.

In words: the top-up estimator is a fully classical object—efficient, regular, doubly robust for the causal target, with no assumption about how well the network’s score span approximates the EIFs (Theorem 1, Proposition 3). The plug-in is the adaptive object: it estimates the working parameter, can be strictly more efficient than the nonparametric bound allows for regular estimators, and pays for that with nonregularity when the working submodel is far from the truth (Proposition 1). Everything in Section 5 is an empirical test of this division of labor.

Remark 1 (No-split implementation and scope). Theorem 1 covers the cross-fitted top-up estimator and Proposition 1 the cross-fitted plug-in; the no-split implementation is covered by neither as stated—it is instead the setting of TDA’s own expansion, which would cover it under a P_0 -Donsker condition on the network’s influence-function class that we have not verified for this architecture—and its validity at moderate n is an empirical matter—Section 5 finds nominal behavior at every

sample size examined ($n = 500\text{--}2000$). Two further caveats. Double robustness of the *point estimate* (Proposition 3) does not extend to inference: consistent variance estimation still requires the relevant nuisances, and Section 5 quantifies the failure modes. Superefficiency likewise cuts both ways: narrower intervals when M_n approximates P_0 well, and a plug-in that tracks the *working* parameter—far from the truth—when it does not; the empirical price appears under outcome misspecification, where only the top-up estimators retain valid inference for $\Psi(P_0)$.

5 Simulation study

Design. All results in this section come from one *unified bank*: every estimator runs on identical datasets and, where applicable, identical nuisance fits, with the propensity always fit by a standalone network so that outcome and weight models can be blinded independently. The base (linear) data-generating process: covariates $X \sim N(0, I_{10})$; treatment $A \mid X \sim \text{Bern}(\text{expit}(0.9X_1 - 0.7X_2 + 0.6X_3))$, truncated to $[0.05, 0.95]$; event hazard $h(k \mid a, x) = \text{expit}(-3.1 + 2k/K + 0.7x_1 + 0.5x_2 + 0.4x_3 - 0.3x_4 + a\tau(x))$ with $\tau(x) = -1 + 0.8x_1 + 0.6x_3$ (the treatment effect changes sign across the population); censoring hazard $\text{expit}(-3.6 + 0.5x_1 + 0.4x_2 + 0.4a + 0.3x_5)$, dependent on treatment and on covariates shared with the outcome (coarsening at random holds; unadjusted estimators are biased). Two additional designs stress the network and the weights: a *nonlinear* design (hazards with sin, quadratic, and interaction terms; heterogeneity $\tau(x) = -1 + 0.9 \tanh x_1 + 0.6x_3 \mathbf{1}\{x_2 > 0\}$; nonlinear censoring) and a *near-positivity* design (propensity coefficients doubled, g_0 truncated only at $[0.02, 0.98]$). We use $K = 30$ throughout; $n \in \{500, 1000, 2000\}$ for the linear design and $n = 1000$ otherwise; 100–150 replications per bank (per-cell coverage MC SE $\approx 1.8\text{--}2.2$ points; ≈ 2.8 for the 60-replication ridge banks). Ground truth is computed by Monte Carlo integration over 4×10^5 covariate draws (given each draw the conditional survival is evaluated exactly, so integration error is negligible at the reported precision); the true Δ_{RMST} is 5.45 bins in the linear design. Tables report bias (time-averaged absolute Monte Carlo bias), the Monte Carlo SD of the estimates, the mean estimated SE (so variance-estimation behavior is diagnosable), MSE, pointwise and simultaneous coverage, and interval width.

Estimators. (i) **Naive plug-in**: the initial network’s curves; (ii) **unadjusted Kaplan–Meier** by arm; (iii) **IPTW $\times$ IPCW KM**: weighting-only correction; (iv) **one-step (AIPCW)**: per-timepoint $\hat{\Psi}_j^{\text{plug}} + P_n D_j$ at the initial fit, with its own multiplier band, plus an *isotonized* variant (each arm projected onto monotone curves), which isolates how much any advantage is explained by monotonicity alone; (v) **output-space survival TMLE**, in two forms sharing the nuisances of (iv): the *per-timepoint* form—each of the $2K$ coordinates gets its own one-dimensional logistic hazard fluctuation with clever covariate $H_t(k, x) = -S(t|a, x) / \{S(k|a, x) g_a(x) S_c(k-1|a, x)\}$, Newton-solved to convergence—and a *universal* form in the spirit of van der Laan and Gruber [12], Cai and van der Laan [1], Rytgaard and van der Laan [9]: a single one-dimensional path whose direction at each step is the $P_n D$ -weighted combination of the per-target clever covariates (computed by suffix sums), moving both arms’ hazards along a shared ϵ until all $2K$ EIF equations hold jointly; (vi) **TDSC**: the plug-in with both SE types (full-EIF and working-model), the full-residual **top-up**, and the cross-fitted plug-in and top-up of Section 4; (vii) **causal survival forests** [4], which fit their own nuisances, evaluated as the doubly robust population benefit at horizons $\{5, 10, \dots, 25\}$ (these depend only on the data, which are seed-identical across banks).

Main results (well-specified, $n = 1000$). Table 1 reports the full bank. Adjustment is essential: unadjusted KM’s bias (0.123) is 27-fold the TDSC plug-in’s (0.0045), and the naive plug-in carries

the regularization bias debiasing exists to remove (0.025). Among the debiased estimators, the TDSC plug-in is best on every criterion: MSE 0.00102 versus 0.00158 for the one-step (a 35% reduction on identical nuisances), pointwise coverage 94.8% versus 93.8%, simultaneous coverage 95.0% versus 92.0%, with 6% narrower intervals (0.133 vs 0.142)—11% narrower under the working-model SEs at 93.9%. The MC-SD/mean-SE columns show why: the plug-in’s sampling SD (0.031) sits below the full-EIF SE (0.034)—the superefficiency signature of Proposition 1, priced fairly by the working SE (0.032)—while the one-step’s SD (0.039) slightly exceeds its SE (0.036). The full top-up gives back part of the variance advantage by construction (MSE 0.00122) in exchange for the unrestricted-target guarantee of Theorem 1.

Figure 3 resolves the same comparison across the time grid, where table averages could in principle hide localized failures. They do not: TDSC’s bias advantage is roughly uniform in t (panel a); its sampling SD runs below the one-step’s at every horizon, with the full-EIF SE bounding it from above and the working-model SE pricing it more closely, if slightly optimistically at late horizons (panel b, the superefficiency gap made visible); and its pointwise coverage stays inside the Monte Carlo error band of 95% across the grid, while the universal output-space TMLE decays to 83–87% at late horizons where its shared one-dimensional path is stretched thinnest (panel c). Figure 6 shows one replication for illustration.

Mechanism decomposition. The same-nuisance ordering in Table 1 decomposes the gain along both design axes: one-step (0.00158) $\rightarrow$ output-space TMLE (0.00127 per-timepoint, 0.00128 universal) $\rightarrow$ TDSC (0.00102)—moving from additive correction to TMLE and from output space to weight space each buy roughly half the total 35% improvement, so weight-space targeting is not redundant given TMLE-ness. The isotonized one-step is indistinguishable from the raw one-step (0.00157 vs 0.00158): monotonicity is not the mechanism. The *universal* output-space path matches the per-timepoint version coordinate-for-coordinate while replacing $2K$ fluctuation sequences with one—confirming that universal output-space targeting works, as Cai and van der Laan [1], Rytgaard and van der Laan [9] construct analytically; TDSC’s distinction is that its path falls out of the architecture’s score gradients generically. Causal survival forests, fitting their own nuisances for a harder conditional-effect problem, carry visible regularization bias into this population target (bias 0.018, coverage 89.2% at horizons 5–25); all debiased semiparametric estimators dominate here. Figure 4 summarizes.

Nuisance misspecification: the two-story framework, demonstrated. Table 2 blinds one nuisance set at a time; Figure 5(a) shows the resulting sampling distributions for the RMST difference. Blinding the outcome network to the main confounder and effect modifier explodes the naive plug-in (bias 0.19) and—this is the theoretically predicted part—the *TDSC plug-in* (bias 0.092, coverage 44.2% full-SE / 33.1% working-SE): with a badly wrong working submodel, the plug-in faithfully solves the projected equations and estimates the *working* parameter, which now sits far from $\Psi(P_0)$ (the oracle-bias conditions of Proposition 1 fail). In the figure this is unmistakable: the plug-in’s entire sampling distribution clusters around $\hat{\Delta}_{\text{RMST}} \approx 2.8$, less than a bin’s width of spread but centered two and a half bins from the truth of 5.45—a precise answer to the wrong question. The *full top-up* restores the unrestricted target exactly as Theorem 1 says it must: bias 0.0063, coverage 93.6%, its distribution re-centered on the truth; pairing the top-up point estimate with the one-step’s initial-fit SE reaches 95.2%. The one-step (94.2%) and, with mild undercoverage, the output-space TMLEs (88.7% universal, 89.3% per-timepoint; Table 2 reports the universal form) remain stable.

Crucially, the failure is visible *in-sample*, without knowing the truth. Figure 5(b) plots the

two diagnostics of Section 3 for every replication: the well-specified and outcome-misspecified runs form nearly disjoint clusters, with the mean maximal full-EIF residual—exactly the top-up magnitude—jumping from 0.018 to 0.099 and the projection residual from 0.27 to 0.41. An analyst who sees a large top-up magnitude has direct evidence that the working submodel’s parameter has drifted from the causal target, and should report the top-up estimator. Blinding the weight models instead leaves all EIF-corrected estimators protected through the hazard arm (biases ≈ 0.010 – 0.012) while weighting-only IPTW $\times$ IPCW KM fails (0.076); all corrected estimators undercover mildly (≈ 89 – 90%) through the shared variance estimate—double robustness of the point estimate, not of inference.

Sample-size scaling. Across $n \in \{500, 1000, 2000\}$ (Table 3; Figure 3d) the TDSC plug-in is nominal at every size (94.7%, 94.8%, 95.0% pointwise; 94.0–95.0% simultaneous) with $n \times \text{MSE}$ stable at 1.02–1.08—consistently below the one-step’s 1.42–1.94, whose excess shrinks with n as first-order theory predicts but is still 38% at $n = 2000$ —and $\sqrt{n} \times \text{bias}$ falling (0.18, 0.14, 0.07). The cross-fitted plug-in tracks the no-split plug-in’s MSE at $n \in \{500, 1000\}$ (the sizes with cross-fit arms) but its intervals are markedly conservative (97.7% at $n=1000$, Table 1; 99.2% at $n=500$), the predicted behavior of full- or ridge-working SEs around a superefficient point estimate whose folds are trained on $4n/5$ observations.

Cross-fitting. In the well-specified bank the cross-fitted top-up—the estimator of Theorem 1—is nominal throughout (95.6% pointwise, 97.0% simultaneous, 96% RMST) at a quantified price: intervals 41% wider than the no-split plug-in (0.188 vs 0.133) and MSE 0.00243 vs 0.00122 for the no-split top-up, reflecting fold nuisances and the held-out residual’s sampling noise. Under *outcome misspecification*—the setting the guarantee exists for—the contrast is exactly as theory orders it: the cross-fitted *plug-in* inherits the working-parameter bias (0.080; coverage 70.0% working-SE, 72.0% full-SE, the former tabulated), while the cross-fitted *top-up* stays nominal (94.3% pointwise) where the no-split plug-in fails; its point estimates are noisier than the no-split top-up’s (bias 0.041 vs 0.006, width 0.234 vs 0.176), so the practical ranking under misspecification is: any full-residual top-up first, cross-fitted for the theorem-backed guarantee, no-split for finite-sample sharpness. At $n = 500$ cross-fitting is disproportionately expensive (top-up MSE 0.00731, width 0.333) because each fold trains on 400 observations.

Robustness: nonlinear design, near-positivity, and the ridge. On the nonlinear design the full ordering is preserved—one-step 0.00203, universal output-space TMLE 0.00158, TDSC plug-in 0.00137 at 94.8%/94.7% coverage—so the mechanism is not an artifact of friendly linear nuisances. Near-positivity is the stress case for everyone: with g_0 reaching 0.02, all estimators undercover (one-step 90.2%; output-space TMLE 84.1% universal, 86.1% per-timepoint; TDSC plug-in 92.8% pointwise but 87.0% simultaneous), and the TDSC plug-in retains the best MSE (0.00143, vs 0.00373 one-step). Part of the shortfall is mechanical: the estimation-side truncation $\hat{g} \in [0.05, 0.95]$ (Appendix A) was left unchanged in this bank while g_0 reaches 0.02, capping weights at 20 where 50 would be correct—an incompatibility with the consistency clause of Assumption 2 that adds truncation bias on top of the variance degradation from heavy IPCW weights; we flag this regime as one where none of the inference here should be fully trusted without weight-stabilization. Finally, the ridge: varying the relative penalty across two orders of magnitude, $\lambda \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$, moves the plug-in’s MSE only within 0.00093–0.00106 and coverage within 93.9–95.3%, while the projection residual grows from 0.18 to 0.40 exactly as shrinkage theory predicts—the ridge is statistically active in principle (Section 4) and empirically inert over this range.

Table 1: Well-specified linear design, benefit curve ($n = 1000$, $K = 30$, 100 replications; coverage MC SE $\approx 2\text{pp}$). MC SD is the Monte Carlo SD of the estimates; mean SE the average estimated standard error. RMST-difference bias/coverage in the final columns.

	Bias	MC SD	mean SE	MSE	Cov.	Width	Simult.	RMST bias	cov
Naive neural plug-in	0.0252	0.051	—	0.00352	—	—	—	−0.52	—
Unadjusted KM	0.1225	0.030	—	0.01665	—	—	—	−3.68	—
IPTW×IPCW KM	0.0132	0.044	—	0.00218	—	—	—	+0.39	—
One-step (AIPCW)	0.0063	0.039	0.036	0.00158	93.8%	0.142	92.0%	−0.14	94%
One-step, isotonized	0.0063	0.039	—	0.00157	—	—	—	−0.14	—
Output-space TMLE (per- t)	0.0042	0.035	0.034	0.00127	92.5%	0.134	—	+0.13	93%
Output-space TMLE (univ.)	0.0059	0.035	0.033	0.00128	91.7%	0.131	91.0%	−0.07	95%
TDSC plug-in (full-EIF SE)	0.0045	0.031	0.034	0.00102	94.8%	0.133	95.0%	−0.04	97%
TDSC plug-in (working SE)	0.0045	0.031	0.032	0.00102	93.9%	0.126	94.0%	−0.04	95%
TDSC + full top-up	0.0044	0.034	0.034	0.00122	93.5%	0.133	—	−0.05	96%
CF plug-in (working SE)	0.0037	0.031	0.039	0.00098	97.7%	0.155	—	−0.03	97%
CF top-up	0.0080	0.048	0.048	0.00243	95.6%	0.188	97.0%	−0.22	96%

Table 2: Nuisance-misspecification grid, benefit curve ($n = 1000$, $K = 30$, 100/150 replications). *Outcome mis.*: outcome hazards blinded to x_1, x_3 ; *weights mis.*: propensity and censoring models blinded to x_1, x_2 . The safeguard row pairs the top-up point estimate with the one-step’s initial-fit SE.

Scenario	Estimator	Bias	MSE	Cov.	Width	Simult.
outcome mis.	naive plug-in	0.1904	0.04028	—	—	—
	one-step (AIPCW)	0.0107	0.00329	94.2%	0.193	95.0%
	output-space TMLE (universal)	0.0155	0.00226	88.7%	0.148	86.0%
	TDSC plug-in (full-EIF SE)	0.0915	0.01004	44.2%	0.176	31.0%
	TDSC plug-in (working SE)	0.0915	0.01004	33.1%	0.150	15.0%
	TDSC + full top-up	0.0063	0.00256	93.6%	0.176	—
	+ one-step SE (safeguard)	0.0063	0.00256	95.2%	0.193	—
	CF plug-in (working SE)	0.0802	0.00774	70.0%	0.217	—
	CF top-up	0.0410	0.00587	94.3%	0.234	98.0%
weights mis.	IPTW×IPCW KM	0.0760	0.00720	—	—	—
	one-step (AIPCW)	0.0114	0.00111	89.4%	0.111	84.7%
	output-space TMLE (universal)	0.0116	0.00118	88.7%	0.110	84.0%
	TDSC plug-in (full-EIF SE)	0.0096	0.00108	90.1%	0.110	87.3%
	TDSC + full top-up	0.0103	0.00113	89.3%	0.110	—

Convergence diagnostics, honestly. The strict projected tolerance $\widehat{\text{sd}}(\widehat{D}_j^{w,\lambda})/(\sqrt{n} \log n)$ is formally met in only 0–13% of replications before the 50-iteration cap binds; the projected means nonetheless end small, coverage is nominal in every well-specified configuration, and the residual top-up removes any dependence on formal convergence. The two recorded diagnostics carry real signal: under outcome misspecification the maximal full-EIF residual jumps more than five-fold ($0.018 \rightarrow 0.099$) and the projection residual by half ($0.27 \rightarrow 0.41$), giving an in-sample warning precisely where the plug-in’s working target departs from the truth (Figure 5b).

Table 3: Sample-size scaling and robustness designs, benefit curve ($K = 30$). Top: linear design across n . Bottom: nonlinear and near-positivity designs at $n = 1000$ (MSE with pointwise coverage in parentheses; TDSC simultaneous coverage in the final column), and the ridge-sensitivity range for the TDSC plug-in ($n = 1000$, 60 replications per λ).

n	Estimator	$n \times \text{MSE}$	Cov.	Width	Simult.
500	one-step	1.94	94.0%	0.218	91.0%
	TDSC plug-in	1.08	94.7%	0.191	94.0%
	TDSC + top-up	1.18	94.0%	0.191	—
1000	one-step	1.58	93.8%	0.142	92.0%
	TDSC plug-in	1.02	94.8%	0.133	95.0%
	TDSC + top-up	1.22	93.5%	0.133	—
2000	one-step	1.42	92.7%	0.093	90.0%
	TDSC plug-in	1.03	95.0%	0.091	94.0%
	TDSC + top-up	1.29	92.6%	0.091	—

Design ($n=1000$)	one-step	univ. osTMLE	TDSC plug-in	TDSC simult.
Nonlinear	0.00203 (93.2%)	0.00158 (92.3%)	0.00137 (94.8%)	94.7%
Near-positivity	0.00373 (90.2%)	0.00245 (84.1%)	0.00143 (92.8%)	87.0%

<i>Ridge sensitivity</i> , TDSC plug-in, $\lambda \in \{0.001, \dots, 0.1\}$: MSE 0.00093–0.00106; coverage 93.9–95.3%; projection residual 0.18→0.40				
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6 Discussion

TDSC turns a two-headed neural survival network into a valid estimator of population treatment benefit at the cost of a few ridge regressions after training, and the simulation program gives its two estimators an honest division of labor. The *plug-in* is the adaptive-efficiency instrument: nominal coverage with the best MSE at every sample size examined, on nonlinear nuisances, and across two orders of magnitude of ridge penalty—but it is inference for the *working* parameter, and under a badly wrong outcome model it tracks that parameter faithfully right past the truth. The *full top-up* is the insurance instrument: a provably regular, doubly robust estimator of the unrestricted target that stayed nominal under outcome misspecification—the setting that breaks the plug-in—and matched the other corrected estimators’ mild undercoverage under weight misspecification (a shared variance-estimation limit, not a point-estimate failure). When the theorem matters, cross-fit it; when finite-sample sharpness matters, the no-split form was nominal at every configuration we examined.

The choice between the two need not be blind. The in-sample diagnostics moved sharply in exactly the settings where the plug-in failed—top-up magnitude five-fold, projection residual by half, with nearly disjoint per-replication clusters (Figure 5b)—so a practical protocol suggests itself: fit both, and let the top-up magnitude arbitrate. When it is small, the plug-in’s working parameter is close to the causal target and its sharper intervals are trustworthy; when it is large, report the top-up.

Limitations and next steps. Near-positivity remains the shared hard regime—every IF-based variance we examined degraded there—and weight stabilization for the targeting step is open. We have not yet mapped sensitivity to the targeting-layer choice, weight-truncation levels, or the grid size K . And two extensions are natural: *calibration-for-benefit*—targeting arm-specific curves within strata of the network’s own predicted benefit, the estimand used to validate individualized-benefit models against randomized trials—and application on frozen pretrained encoders (e.g. pathology

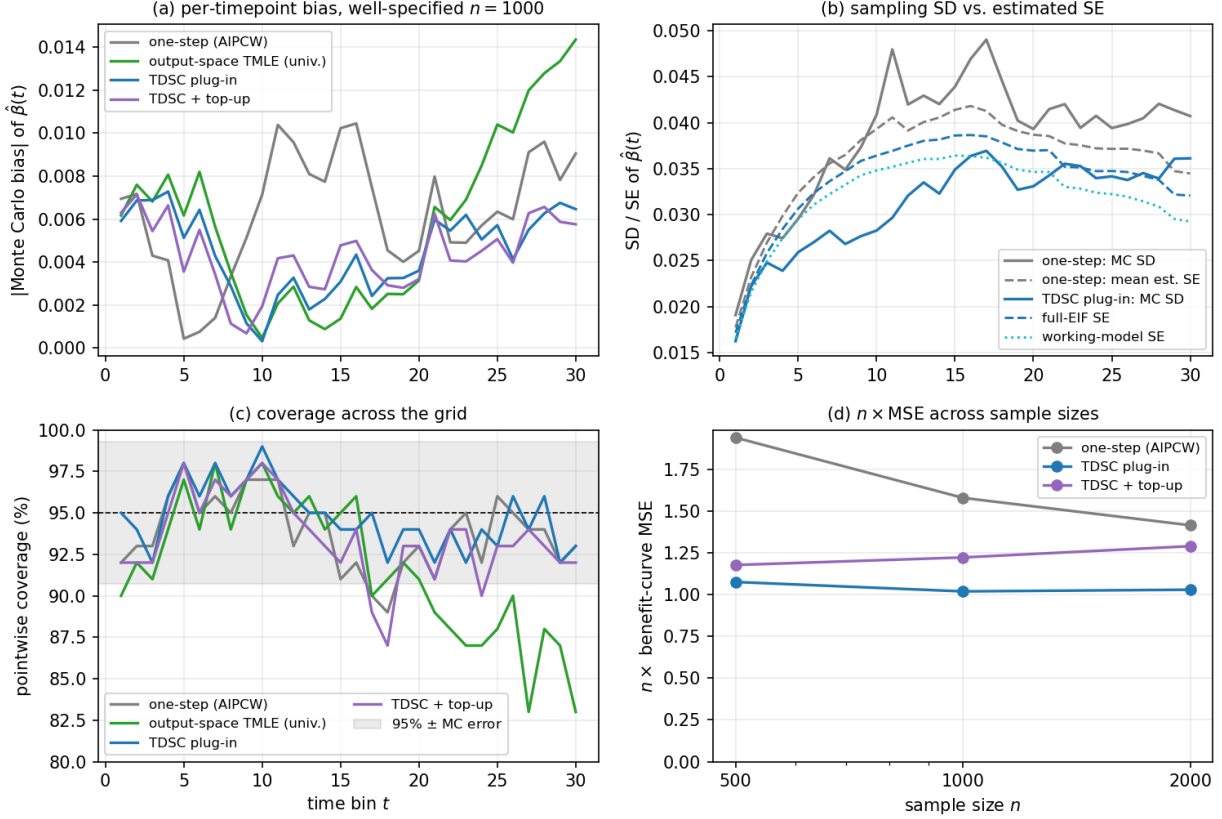


Figure 3: Estimator behavior resolved across the time grid (well-specified linear design, 100 replications). (a) Per-timepoint absolute Monte Carlo bias of $\hat{\beta}(t)$ at $n = 1000$: TDSC’s advantage is roughly uniform in t . (b) Sampling SD versus mean estimated SE: the TDSC plug-in’s SD (solid blue) runs below the one-step’s (solid gray) at every horizon; the full-EIF SE (dashed) bounds it from above—the superefficiency gap of Proposition 1—while the working-model SE (dotted) prices it more closely. (c) Pointwise coverage across the grid with the 95% Monte Carlo error band (shaded): TDSC stays inside the band throughout; the universal output-space TMLE degrades at late horizons. (d) $n \times \text{MSE}$ across $n \in \{500, 1000, 2000\}$: flat near 1.0 for the TDSC plug-in, converging from above for the one-step.

foundation models), which the last-layer construction supports directly.

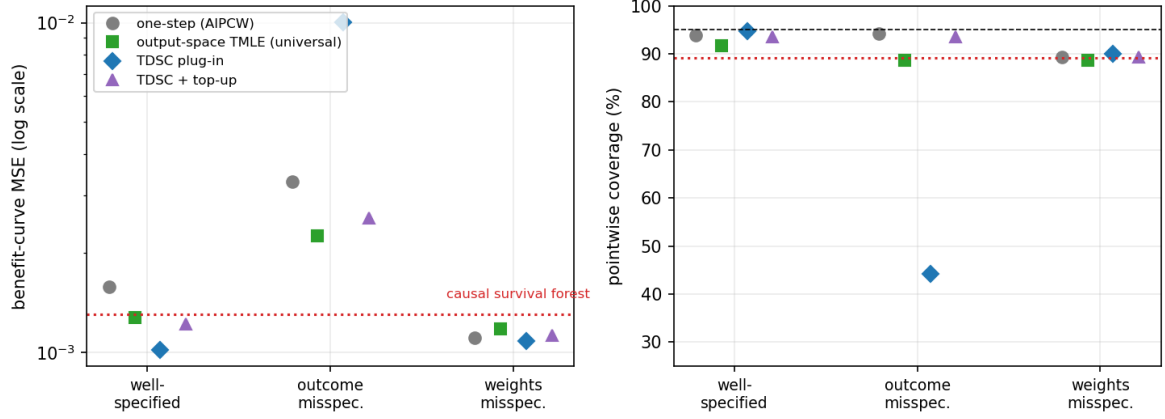


Figure 4: Head-to-head comparison across nuisance scenarios (benefit-curve MSE, log scale; pointwise coverage, 95% dashed): one-step, universal output-space TMLE, TDSC plug-in, and TDSC top-up on identical nuisances, with the causal survival forest (own nuisances, horizons 5–25) as the dotted reference. Well-specified, the estimators order one-step $\rightarrow$ output-space TMLE $\rightarrow$ TDSC; under outcome misspecification the plug-in is the most exposed—it estimates the working parameter—while the top-up and one-step remain nominal.

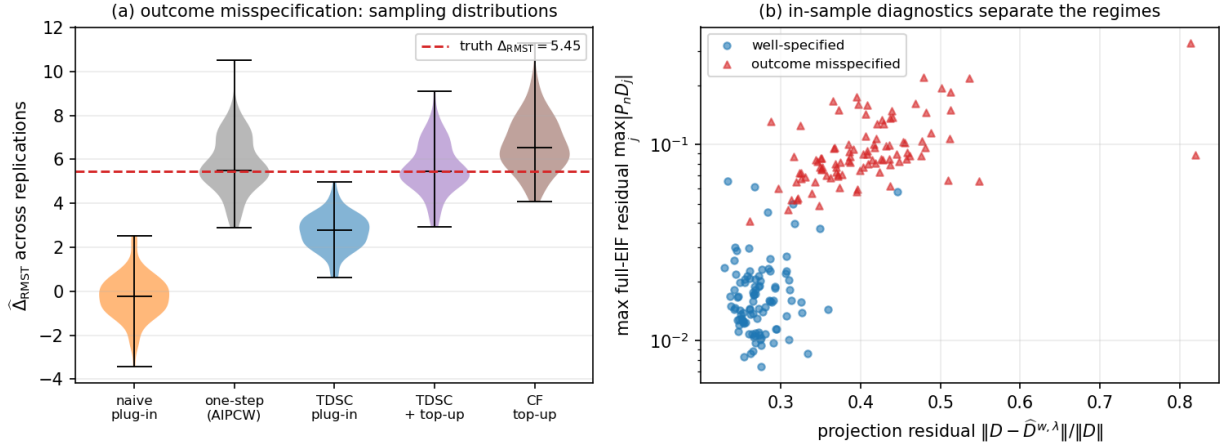


Figure 5: The two-estimator story under outcome misspecification ($n = 1000$, 100 replications). (a) Sampling distributions of $\hat{\Delta}_{\text{RMST}}$: the TDSC plug-in concentrates tightly around its *working* parameter (≈ 2.8), far from the truth of 5.45 (dashed), while the full top-up and the cross-fitted top-up re-center on the truth; the naive plug-in is worst. (b) The two in-sample diagnostics, one point per replication: well-specified runs (circles) and outcome-misspecified runs (triangles) form nearly disjoint clusters in (projection residual, max full-EIF residual) space, so the regime is identifiable from the fitted object alone—a large top-up magnitude is the signal to report the top-up estimator.

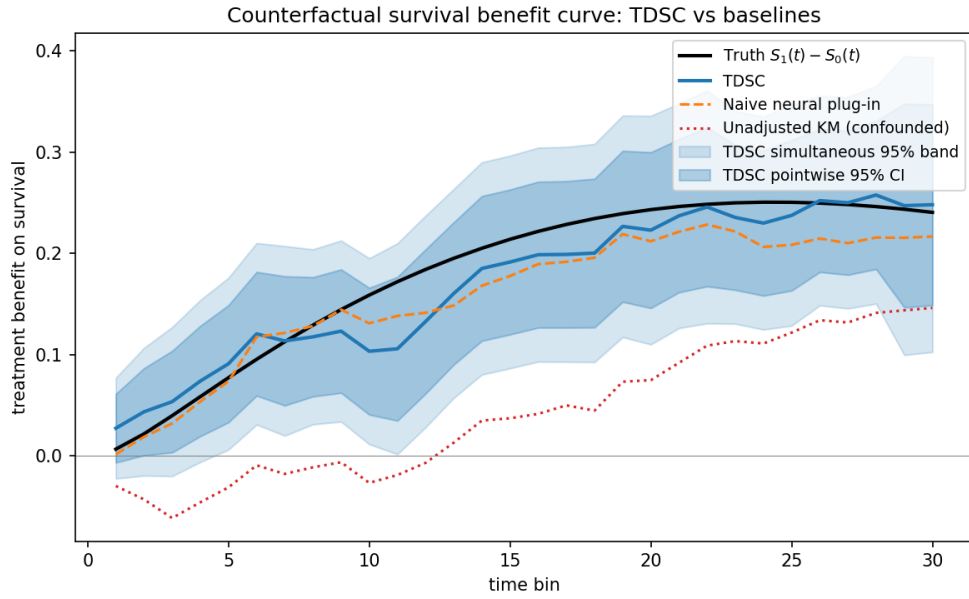


Figure 6: One replication ($n = 1000$, seed 7), shown for illustration. The TDSC benefit curve tracks the truth, which lies inside both the pointwise CI and the simultaneous band; the naive plug-in drifts low at later times, and unadjusted KM is severely biased (wrong-signed at early times).

A Reproducibility

All code, raw Monte Carlo stores, and the exact scripts behind every table are at <https://github.com/blind-contours/tdsc>. Networks: trunk $d \rightarrow 64 \rightarrow 64$ (ELU); propensity head $64 \rightarrow 64 \rightarrow 1$; per-arm outcome heads $64 \rightarrow 64 \rightarrow K$; censoring network $(d+1) \rightarrow 64 \rightarrow 64 \rightarrow (K-1)$. Training: full-batch Adam, lr 10^{-3} , max 500 epochs, early stopping (patience 25) on a 20% validation split, propensity loss weight 1.0. Targeting: ridge scale $\lambda = 0.01 \times \text{mean diag}(G^\top G)$ (a finite-sample stabilizer; the theory requires only that the relative ridge vanish, Section 4); $\gamma_0 = 1$ with up to 12 halvings; max 50 iterations; stopping tolerance $\widehat{\text{sd}}(\widehat{D}_j^{w,\lambda})/(\sqrt{n} \log n)$ on the projected means. Truncation: $\hat{g} \in [0.05, 0.95]$, $\hat{S}_c \geq 0.05$, conditional survival floored at 10^{-3} inside (1); these levels were used in *all* banks, including near-positivity, where g_0 reaches 0.02 (see Section 5 for the induced truncation bias). Bands: 1000 Gaussian-multiplier draws. Cross-fitting: $V = 5$, fold networks trained and targeted on complements with the same hyperparameters. Output-space TMLE, per-timepoint form: per-coordinate logistic fluctuations solved by Newton steps (ϵ clipped to $[-5, 5]$), recomputing the clever covariate each round, up to 25 rounds per coordinate; universal form: shared one-dimensional path along the $P_n D$ -weighted suffix-sum clever covariate, ϵ steps clipped to $[-0.5, 0.5]$, up to 200 steps (70–105 observed), same $\text{sd}/(\sqrt{n} \log n)$ tolerance. Causal survival forests: `grf` 2.6.1, `causal_survival_forest` with `target = "survival.probability"`, 1,000 trees, default tuning, one forest per horizon $\{5, 10, 15, 20, 25\}$, population effect and SE from `average_treatment_effect`. Every bank uses a standalone propensity network and paired data seeds; replications per bank: 100 (well $n=1000$, outcome-misspecified, near-positivity, $n=500$, $n=2000$), 150 (weights-misspecified, nonlinear), 60 per ridge value. Truth: Monte Carlo integration over 4×10^5 covariate draws.

B Proof sketches

Theorem 1. All influence functions in this proof are the nonparametric EIFs (1); superscripts are omitted. Per coordinate, since $\widehat{\Psi}_j^+ = \Psi_j(\hat{P}) + P_n \widehat{D}_j$ and the EIF has the form $\widehat{D}_j = \widehat{D}_j^{\text{mart}} + \hat{S}_j - \Psi_j(\hat{P})$, the standard one-step decomposition holds exactly:

$$\widehat{\Psi}_j^+ - \Psi_j(P_0) = P_n D_{0,j} + (P_n - P_0)(\widehat{D}_j - D_{0,j}) + R_{2,j}, \quad R_{2,j} = P_0 \widehat{D}_j + \Psi_j(\hat{P}) - \Psi_j(P_0).$$

For the remainder, the survival EIF is exact to second order: writing $S_h(k) = \prod_{j \leq k} (1 - h(j))$ for the conditional survival under a hazard h , the telescoping identity $S_{\hat{h}}(t) - S_{h_0}(t) = \sum_{k \leq t} S_{h_0}(k-1) (h_0(k) - \hat{h}(k)) S_{\hat{h}}(t)/S_{\hat{h}}(k)$ gives $R_{2,j}$ as a sum over $k \leq t$ of integrals of $\{h_0(k) - \hat{h}(k)\}$ against $\{\hat{g}_a \hat{S}_c(k-1) - g_{0,a} S_{c,0}(k-1)\}/\{\hat{g}_a \hat{S}_c(k-1)\}$ -type differences; under the positivity bounds of Assumption 1, Cauchy–Schwarz gives $|R_{2,j}| \lesssim \|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ by Assumption 2, applied to the targeted per-fold hazards (this display also proves Proposition 3: R_2 vanishes if either factor’s limit is correct). For the empirical-process term, cross-fitting makes the fold estimate $\widehat{D}_{v,j}$ independent of fold v , so conditional on the training folds, $(P_{n,v} - P_0)(\widehat{D}_{v,j} - D_{0,j}) = O_p(n^{-1/2} \|\widehat{D}_{v,j} - D_{0,j}\|_{P_0}) = o_p(n^{-1/2})$ by Chebyshev whenever $\|\widehat{D}_{v,j} - D_{0,j}\|_{P_0} = o_p(1)$, which follows from the individual nuisance consistency in Assumption 2; summing the V folds gives the pooled statement and the coordinate-wise expansion, and joint convergence follows from the multivariate CLT applied to the stacked D_0 . For the variance estimator: the positivity bounds of Assumption 1 give a bounded envelope for the estimated EIFs, which with $\|\widehat{D}_j - D_{0,j}\|_{P_0} = o_p(1)$ and out-of-fold evaluation yields $P_n[\widehat{D} \widehat{D}^\top] \rightarrow_p \Sigma_0$, so the reported standard errors and the band’s $\widehat{\text{se}}_t$ are consistent. Proposition 1 follows the working-parameter and original-parameter arguments

of Li et al. [7] with two modifications: the linear expansion is evaluated out-of-fold, and their Donsker-type condition is imported on the training folds, where the projected score equations are solved. Superefficiency follows from the likelihood-based oracle-submodel extension of those results (the loss here is a per-bin Bernoulli likelihood) together with van der Laan et al. [14].

Corollary 1 and Proposition 2. Both maps are bounded linear in Ψ , so asymptotic linearity transfers with the transformed influence functions. Given joint normality with consistently estimated Σ_0 (IF second moments; see the Theorem 1 sketch) and the nondegeneracy $\min_t \sigma_t > 0$, the conditional multiplier process $n^{-1/2} \sum_i \xi_i \{\widehat{D}_{\beta,t}(O_i) - \bar{D}_{\beta,t}\} / \widehat{\text{se}}_t$ —the studentized, centered statistic the algorithm computes; the centering constant is $O_p(n^{-1/2})$, hence asymptotically negligible—converges weakly to the standardized Gaussian limit (conditional multiplier CLT), whose sup-distribution is continuous and strictly increasing at its 0.95 quantile, so the estimated sup-quantile is consistent; the $c^* \geq 1.96$ floor only enlarges the band.

Proposition 3. By definition $\widehat{\Psi}_j^+ = P_n[\hat{S}(t \mid a, X)] + P_n \widehat{D}_j$, which is precisely the AIPCW one-step at nuisances $(\hat{h}^{\text{targ}}, \hat{g}, \hat{S}_c)$. Its population bias is the R_2 term above evaluated at those limits, which vanishes under either-arm consistency. $\square$

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